

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Contact cleaner, Part Number 3824510, or equivalent
- Lubriplate™ 105, Part Number 3163086, or equivalent

Additional Service Items

- Dial indicator
- Hydraulic press capable of 2224 N [5,000 lbf]

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

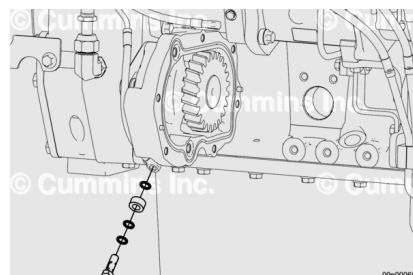
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-018.html>)
- Remove the low-temperature aftercooler (LTA) water pump support bracket. Refer to Procedure 008-195 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-195.html>)

- Remove the LTA water pump. Refer to Procedure 008-109 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-109.html)
- Remove the LTA water pump drive. Refer to Procedure 009-092 in Section 9.
(/qs3/pubsys2/xml/en/procedures/57/57-009-092.html)

Remove

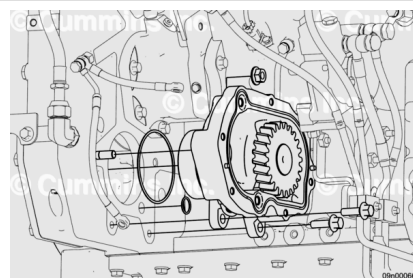
Disconnect lubricating oil supply line.

Remove and discard sealing washers.



Remove offset accessory drive from gear housing.

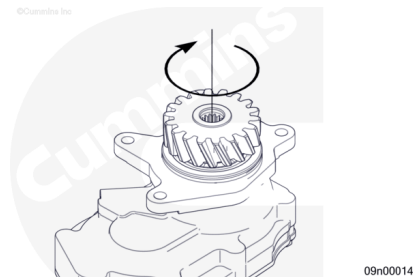
Remove and discard sealing o-ring seal and lubricating oil passage o-ring seal.



Inspect

Turn gear of offset accessory drive.

Rebuild drive if it does **not** turn smoothly.

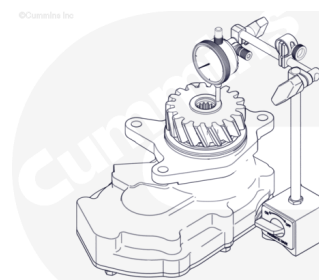


Measure end clearance.

Offset Accessory Drive End Clearance		
mm		in
0.33	MIN	0.013

Offset Accessory Drive End Clearance

mm		in
0.45	MAX	0.018

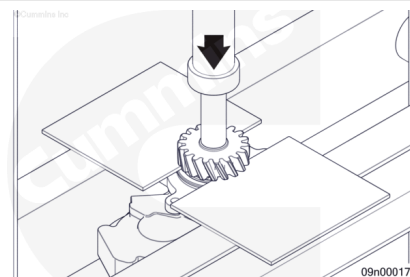


09n00015

Replace thrust bearings if clearance is **not** within specifications.

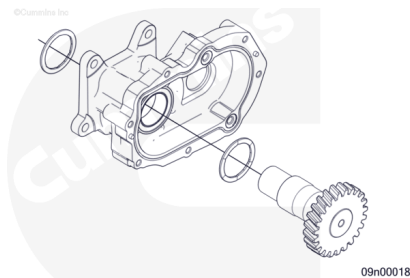
Disassemble

Use hydraulic press to remove gear from offset accessory drive.



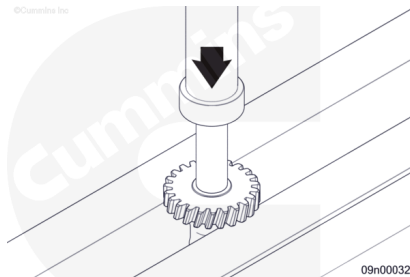
09n00017

Remove shaft assembly and thrust bearings from offset accessory drive.



09n00018

Use hydraulic press to press shaft out of remaining gear.



09n00032

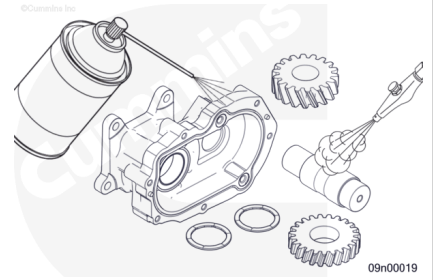
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris or dirt can cause personal injury.



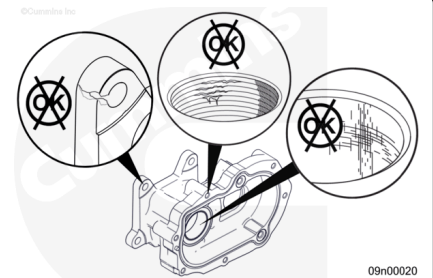
Clean components. Use contact cleaner, Part Number 3824510, or equivalent.

Dry with compressed air.

Inspect offset accessory drive for:

- Cracks
- Damaged sealing surfaces
- Damaged threads
- Damaged bearings

Replace any damaged components.

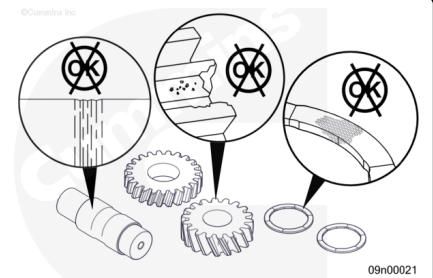


Inspect gears for damaged or missing teeth.

Inspect shaft for damage and scratches or scoring on bearing mating surfaces.

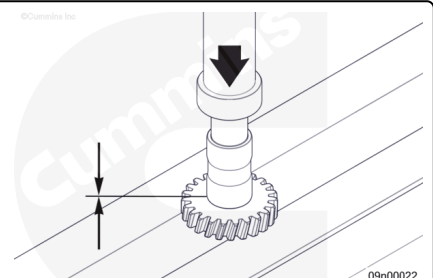
Inspect thrust bearings for excessive wear.

Replace any damaged or worn components.



Assemble

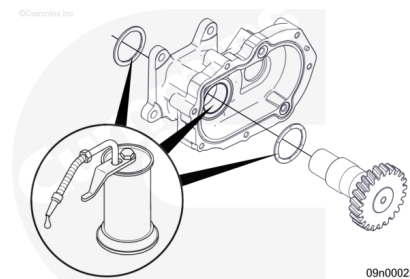
Use hydraulic press to press shaft into rear gear until shoulder on the shaft contacts gear.



Install thrust bearings into offset accessory drive. The grooved side of the bearings face away from offset accessory drive.

Lubricate thrust bearings and housing bearings with clean engine lubricating oil.

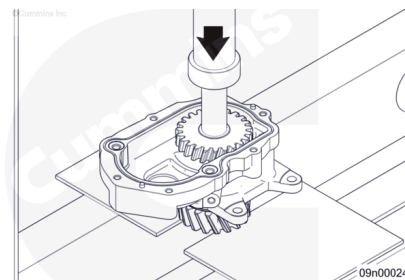
Install shaft assembly into rear of offset accessory drive.



⚠ CAUTION ⚠

Do not use drive housing as a support to press in shaft assembly. Damage to drive housing can result.

Use hydraulic press to press shaft assembly into front gear until shoulder on the shaft assembly contacts gear.



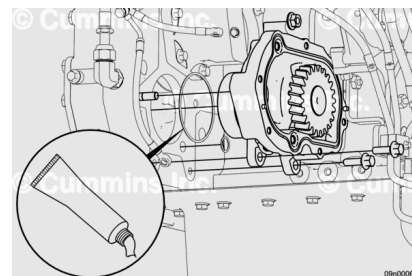
Install

Install new o-ring seals.

Lubricate radial o-ring seal with Lubriplate™ 105, Part Number 3163086, or equivalent.

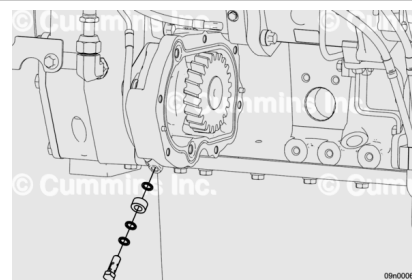
Install offset accessory drive.

Torque Value: 113 n•m [83 ft-lb]



Connect lubricating oil supply line. Use new sealing washers.

Torque Value: 15 n•m [133 in-lb]



Finishing Steps

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- Install the LTA water pump drive. Refer to Procedure 009-092 in Section 9.
(/qs3/pubsys2/xml/en/procedures/57/57-009-092.html)
- Install the LTA water pump. Refer to Procedure 008-109 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-109.html)
- Install the LTA water pump support bracket. Refer to Procedure 008-195 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-195.html)
- Fill the LTA cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-018.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate engine. Check for leaks.